computer vision

CO2 AT3 — class Task

Code : ITA0502

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1. Grey-level Transformations using log and Gamma Correction.

Grey-level transformations improve the brightness and contrast of an image. log transformation and Power-law (gamma) transformation are suitable for enhacing dark regions.

1. log transformation.

The formula is:

$$S = c \log(1+r)$$

where:

- r = input pixel intensity.
- S = Output pixel intensity.
- c = Scalling Constant.

Processing steps:

1. Read the dark image.
2. Convert pixel intensities to floating-point values.
3. Apply the logarithmic transformation to each pixel.
4. Normalize the output intensity values to the range 0-255.
5. Display the enhanced image.

2. Gamma Correction.

The formula is:

$$S = cr^{\gamma}$$

for normalized pixel values b/w 0 and 1, choose $\gamma < 1$, such as $\gamma = 0.5$ to brighten the image.

Conclusion: Both methods improve the visibility of dark exhibits. Gamma Correction provides flexible brightness Control, while log transformation strongly expands darker intensity values.

2) Histogram Equalization for fingerprint Enhancement.

Histogram equalization improves image Contrast by redistributing pixel intensities over a wider range.

$$S_k = (L-1)\sum_{j=0}^{k} P(r_j)$$

Where:

- L = no. of intensity levels, usually 256.
- $P(r_j)$ = normalized histogram probability.
- S_k = mapped output intensity.

Processing steps:

1. Read the fingerprint image.
2. Convert it to grayscale if necessary.
3. Calculate the histogram of pixel intensities
4. Normalize the histogram to obtain probabilities
5. Calculate the Cumulative distribution function.
6. Display the enhanced fingerprint.

3) Salt-and-pepper Noise Removal using a Median filter.

A median filter is a nonlinear spatial filtering technique used to remove salt-and-pepper noise while preserving object edges.

for a 3x3 neighborhood, the output pixel is;

$$g(x,y) = \text{median}\{f(s,t)\}$$

where $f(s,t)$ represents the pixel intensities in the neighborhood.

Ex: for the neighborhood.

$$\begin{bmatrix} 10 & 12 & 11 \\ 13 & 255 & 14 \\ 12 & 11 & 10 \end{bmatrix}$$

the sorted values are

10, 10, 11, 11, 12, 12, 13, 14, 255

the Median is 12.

So the noisy centre pixel 255 is replaced by 12.

4. Image Sharpening using Laplacian and Unsharp masking.

Image Sharpening enhances fine details and edges by emphasizing rapid changes in intensity.

method 1: Laplacian Sharpening.

laplacian is a second-order derivative operator:

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

A common discreate laplacian mask is:

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

the sharpend image can be obtained using:

$$g(x,y) = f(x,y) + \nabla^2 f(x,y)$$

Method 2: unsharp masking.

The formula is:

$$g = f + k(f - \bar{f})$$

Where:

- f = original image
- $\bar{f}$ = blurred image
- $f - \bar{f}$ = unsharp mask
- k = sharpening strength.

5) Frequency-Domain low-pass filtering.

A low-pass filter smooths an image by allowing low-frequency components to pass while attenuating high-frequency components.

$$G(u,v) = H(u,v)\, F(u,v)$$

where:

- $F(u,v)$ = fourier transform of the input image.
- $H(u,v)$ = low-pass filter transfer function.
- $G(u,v)$ = filtered frequency spectrum.

1. Ideal low-pass filter (ILPF)

passes frequencies inside the cutoff radius and rejects frequencies outside it. It can produce ringing artifacts.

2. Butterworth low-pass filter (BLPF)

provides a smoother transition b/w passed and rejected frequencies. Its order controls the sharpness of the transition.

3. Gaussian low-pass filter (GLPF)
Attenuates high frequencies smoothly and reduces ringing artifacts.

6. Frequency-Domain High-pass (or) High-boost filtering.
A high-pass filter enhances high-frequency components, such as sharp edges and fine details. It can improve the visibility of thin blood vessels in a retinal image.

$$G(u,v) = H(u,v)F(u,v)$$

$$H(u,v) = \begin{cases} 0, & D(u,v) \le D_0 \\ 1, & D(u,v) > D_0 \end{cases}$$

where $D(u,v)$ is the distance from the frequency origin and D_0 is the cutoff frequency.

High-boost filtering:

$$g(x,y) = A f(x,y) - f_L(x,y)$$

Processing steps:
1. Read the retinal fundus image.
2. Compute its fourier transform.
3. Apply a high-pass or high-boost filter.
4. Compute the inverse fourier transform.
5. Normalize the result (transform) for display.

7. Point Detection and Line Detection.

Point detection identifies isolated pixels that differ significantly from their surroundings.

1. Point detection mask.

A common mask is:

$$w = \begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

$$R(x,y) = w * f(x,y)$$

if $|R(x,y)| > t$, the pixel is classified as a possible isolated point.

2. Line detection Masks.

Horizontal line mask:

$$w_H = \begin{bmatrix} -1 & -1 & -1 \\ 2 & 2 & 2 \\ -1 & -1 & -1 \end{bmatrix}$$

Vertical line Mask:

$$w_V = \begin{bmatrix} -1 & 2 & -1 \\ -1 & 2 & -1 \\ -1 & 2 & -1 \end{bmatrix}$$

A diagonal mask for one diagonal orientation

$$w_D = \begin{bmatrix} -1 & -1 & 2 \\ -1 & 2 & -1 \\ 2 & -1 & -1 \end{bmatrix}$$

Processing Steps:

1. Convert the bottle image to grayscale.
2. Apply the point detection mask to locate possible bubbles.
3. Apply horizontal, vertical, and diagonal line masks to identify cracks.
4. Calculate the absolute filter responses.
5. Compare responses against a threshold.
6. Mark detected defects on the original image.

8) Image Thresholding for Handwritten Answer Sheets.

Thresholding converts a grayscale image into a binary image, separating foreground handwriting from the background.

1. Global thresholding.

A single threshold T is applied to the entire image:

$$g(x,y) = \begin{cases} 0, & f(x,y) < T \\ 255, & f(x,y) \geq T \end{cases}$$

2. Adaptive thresholding.

Adaptive thresholding calculates a local threshold for each pixel using the surrounding neighborhood.

$$T(x,y) = \text{Local Mean}(x,y) - C$$

where C is a constant.

The output is black if the pixel intensity is below the local threshold; otherwise, it is white.

9) Region Growing for Lake Extraction from a Satellite Image.

Region growing is an image segmentation technique that starts with a seed pixel and adds neighboring pixels that satisfy a similarity criterion.

Similarity criterion:

A simple intensity-based condition is:

$$|I(x,y) - I_s| \leq T$$

- $I(x,y)$ = intensity of the candidate pixel.
- I_s = seed intensity.
- T = intensity tolerance.

10) Edge Linking and Boundary Detection for Mechanical Gears.

Edge linking connects fragmented edge pixels while boundary detection identifies the complete outer contour of an object.

1. Local edge linking.

Connects nearby edge pixels based on:

- Spatial Proximity
- Similar gradient magnitude,
- Similar gradient direction.

2. Hough transform.

The Hough transform detects geometric shapes by mapping edge pixels into a Parameter space.

$$P = x\cos\theta + y\sin\theta$$

for circular gear boundaries, a circular Hough transform can be used:

$$(x-a)^2 + (y-b)^2 = r^2$$

3. Boundary detection.

After linking the edges:

v Convert the linked edges into a suitable binary representation.

2. Apply morphological closing to connect small gaps.

3. fill enclosed regions if required.

4. Extract the external Contour.

5. Overlay the detected boundary on the original gear image.